

Multiple Choice (10 marks)

1. Statement 1| Word 2 Vec parameters were not initialized using a Restricted Boltzman Machine. Statement 2| The tanh function is a nonlinear activation function.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
2. Statement 1| The BLEU metric uses precision, while the ROGUE metric uses recall. Statement 2| Hidden markov models were frequently used to model English sentences.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
3. Averaging the output of multiple decision trees helps __.
 - (a) Increase bias
 - (b) Decrease bias
 - (c) Increase variance
 - (d) Decrease variance
4. Statement 1| Overfitting is more likely when the set of training data is small. Statement 2| Overfitting is more likely when the hypothesis space is small.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
5. The numerical output of a sigmoid node in a neural network:
 - (a) Is unbounded, encompassing all real numbers.
 - (b) Is unbounded, encompassing all integers.
 - (c) Is bounded between 0 and 1.
 - (d) Is bounded between -1 and 1.

True / False (10 marks)

Answer True or False

6. True or False: Calculating $P(H|E, F)$ requires $P(E, F)$, $P(H)$, and $P(E|H)$, $P(F|H)$ without any conditional independence information.
7. True or False: The original ResNets and Transformers are both types of feedforward neural networks.
8. True or False: Fully connected networks have been shown to outperform convolutional networks in classifying high-resolution images.
9. True or False: Linear hard-margin SVM can only be effectively used when the training data is linearly separable.
10. True or False: The variance of the Maximum A Posteriori (MAP) estimate is lower than the variance of the Maximum Likelihood Estimate (MLE).

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the computational cost of performing a single gradient descent update when the gradient of the cost function is represented as a vector. What factors influence this cost?
12. Discuss the implications and methodologies of applying unsupervised learning to analyze a large dataset of medical records from heart disease patients to identify potential treatment clusters.

End of Paper